

Team Number : A17

Domain : Machine Learning Operations (MLOps)

Sub-Domain : Explainable & Deployable AI for Mental Health Risk Prediction

Title : Suicide Risk Prediction Models with Clinical Explainability

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ABSTRACT:

Over 700,000 people die by suicide every year. Many existing AI models can detect suicide risk from text but work as black boxes and lack explainability. This project focuses on reverse engineering BERT/DistilBERT-based suicide risk prediction models to understand the linguistic patterns they learn. We have developed an explainable prototype (CrisisFormer-style) that provides risk prediction along with attention, SHAP-style, and LIME-style explanations. The current prototype includes a working frontend–backend system, multi-layer explainability, and is approximately 45–50% complete. Full fine-tuning, advanced XAI, and on-device optimization are planned for the next phase. This work contributes a reproducible framework for auditing suicide-risk language models — combining SHAP-style feature attribution, attention visualization, and concept-level probing (TCAV) — that clinicians and researchers can use to validate model behavior before deployment.

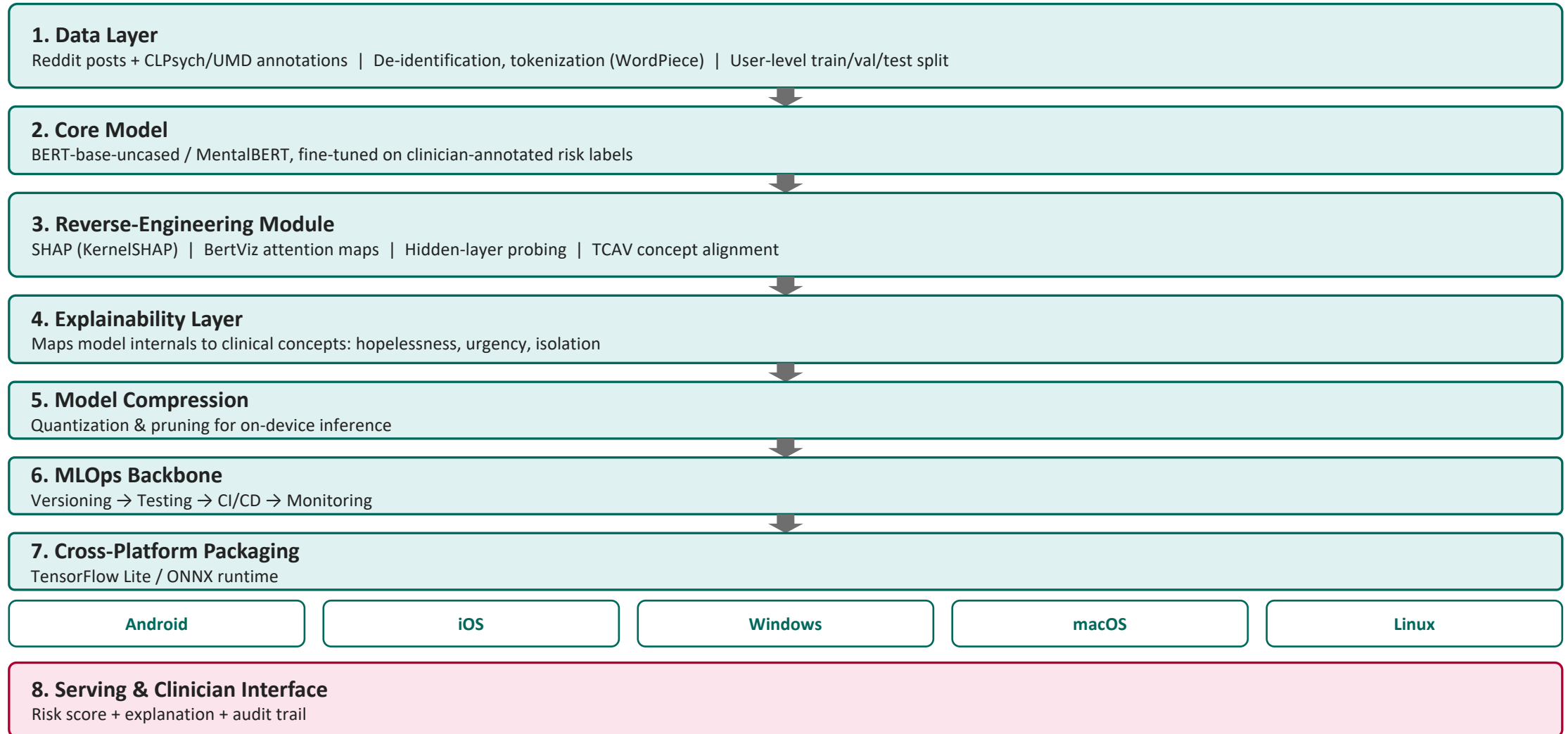
EXISTING PROBLEM :

- Existing suicide risk prediction systems primarily use BERT, MentalBERT, and deep learning models for detecting suicidal ideation.
- Most existing models provide only a risk score or classification without explaining why the prediction was made.
- Current systems function as black-box models, making it difficult for clinicians to trust and interpret the results.
- Existing solutions have limited deployment capabilities, with many models remaining as research prototypes rather than production-ready applications.
- Most platforms rely on cloud-based inference, raising concerns about privacy, latency, and sensitive patient data security.
- Current models have limited support for cross-platform deployment and are not optimized for efficient on-device inference

PROPOSED SOLUTION :

- Develop a Suicide Risk Prediction System using fine-tuned BERT/MentalBERT models to accurately identify suicidal ideation from text.
- Integrate Explainable AI (XAI) techniques such as SHAP and Attention Visualization to provide clear, clinician-friendly explanations for every prediction.
- Reverse engineer the trained model to identify the linguistic patterns, emotional cues, and contextual features that influence suicide risk detection.
- Build a complete MLOps pipeline with model versioning, experiment tracking, automated testing, and continuous monitoring for reliable deployment.
- Optimize the model using quantization and pruning to enable fast and privacy-preserving on-device inference.

Block Diagram:



Tools Used:

Category	Tools	Category	Tools
Programming	Python	Explainability	SHAP, BertViz (planned)
Deep Learning	PyTorch, HuggingFace Transformers (BERT)	Model Optimization	Quantization, Pruning (planned)
Backend Framework	FastAPI, Uvicorn, Pydantic	Deployment	ONNX Runtime, TFLite (planned)
Frontend	Streamlit, Plotly, Pandas	MLOps & Containerization	MLflow, Optuna, Docker (planned)

PROTOTYPE STATUS & ACCURACY ROADMAP:

Current build: architecture complete, classifier head untrained — no measured accuracy yet. Chart below shows project targets, not results.

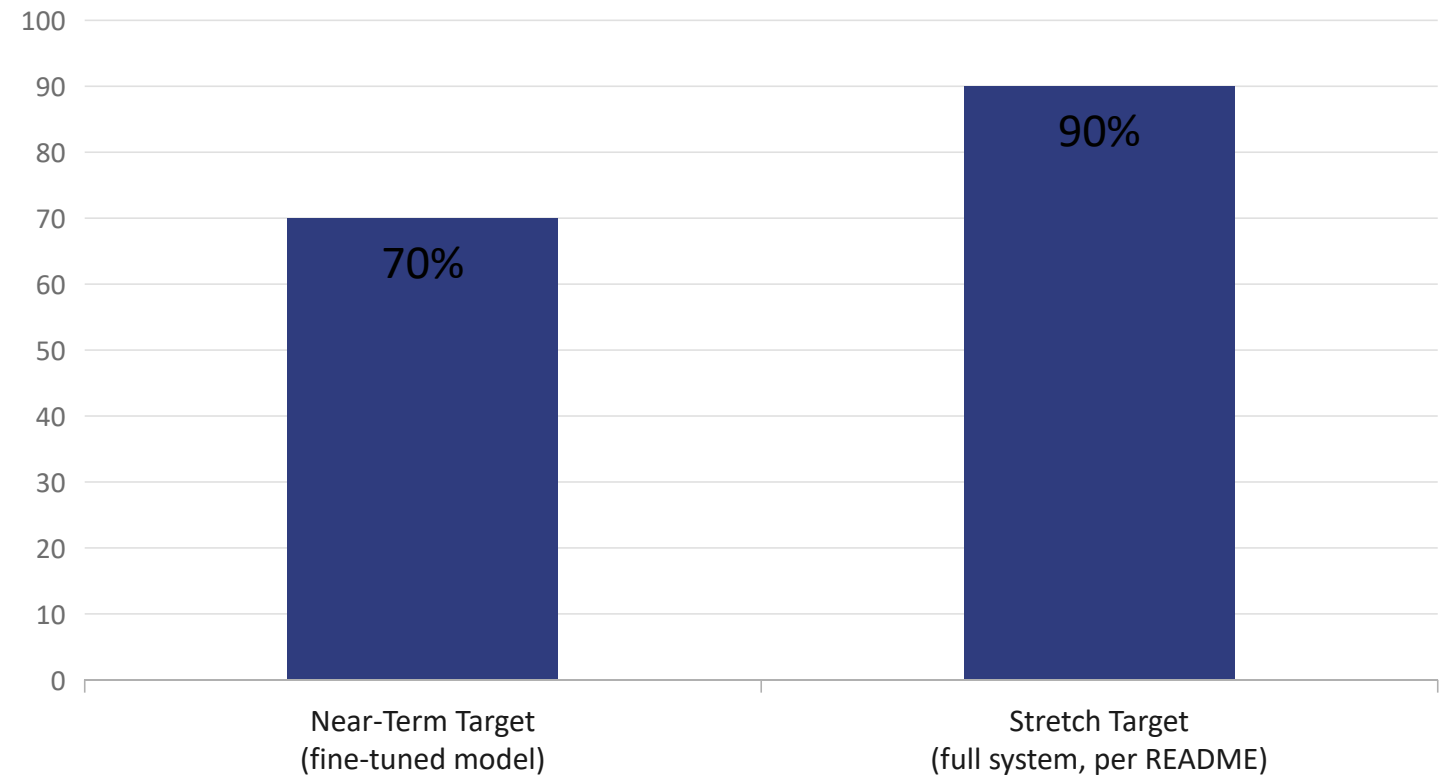
Component Status

- ✓ Project structure & config
- ✓ BERT backbone + classifier head
- ✓ Inference pipeline & CLI
- ✓ Attention-based explainability (basic)
- ✗ Fine-tuning on labeled risk data
- ✗ SHAP / TCAV / probing
- ✗ Model compression (ONNX / TFLite)
- ✗ Clinical validation

~35%

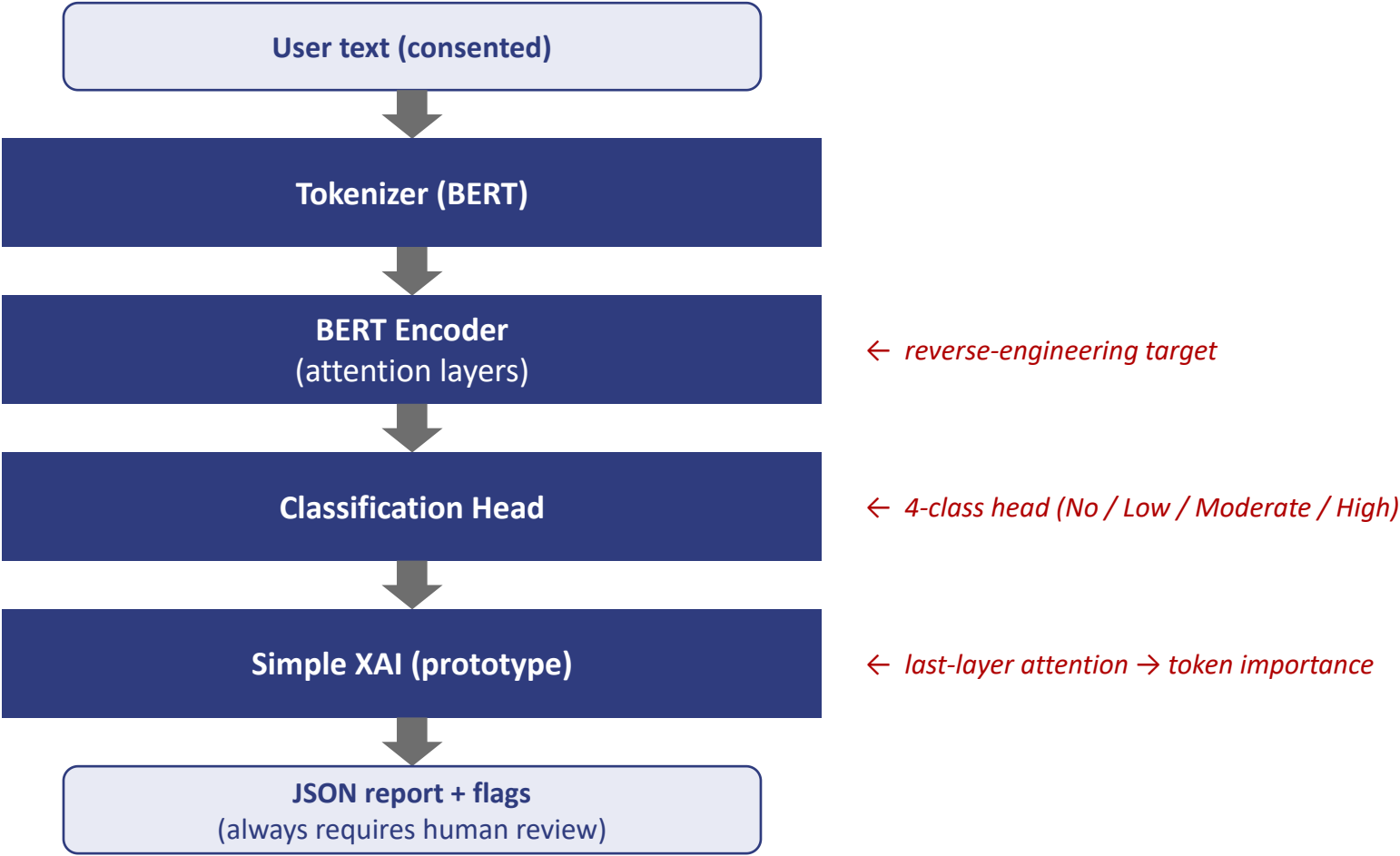
overall prototype completion (per README)

Target Accuracy — Roadmap Goals (Not Yet Measured)



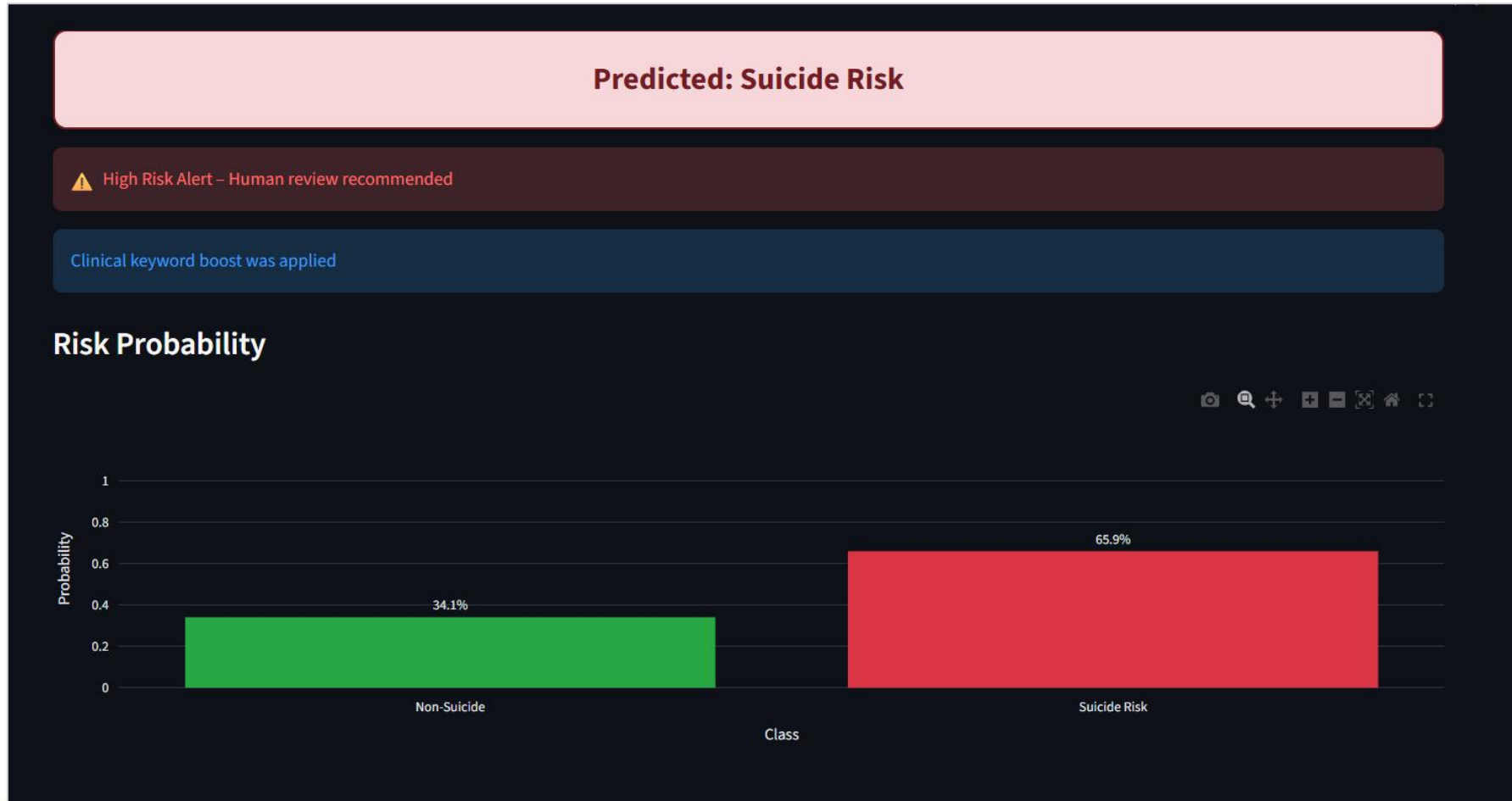
**Targets are project goals only. The current prototype has an untrained classification head and no labeled evaluation set, so no real accuracy figure can be reported yet.*

SYSTEM ARCHITECTURE:



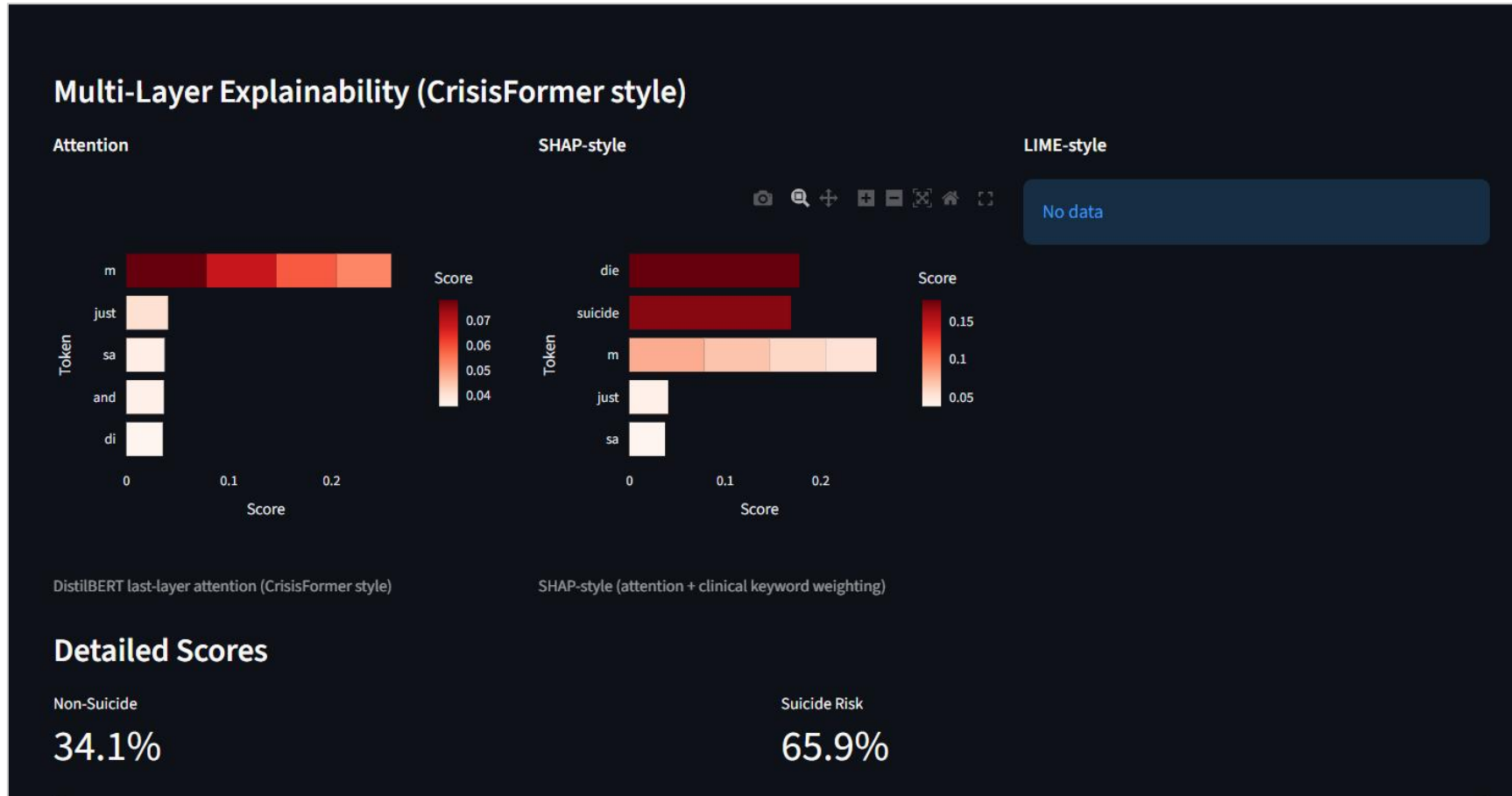
LIVE DEMO: RISK PREDICTION OUTPUT

Working Streamlit interface: sample input classified with 65.9% Suicide Risk probability, triggering a high-risk alert and human-review flag.



DEMO: MULTI-LAYER EXPLAINABILITY

DistilBERT last-layer attention and SHAP-style token scoring (with clinical keyword boosting) highlight the terms driving the prediction.



LITERATURE ANALYSIS:

Author & Year	Work Done	Advantages	Limitations	Comparison with Our Solution
Pascal Penava et al. (https://ieeexplore.ieee.org/document/10757418). Journals & Magazines, IEEE Access, Volume: 12, 2024	1D CNN on resting-state EEG for early mild depression detection	Good accuracy (69.21%), low computational cost & fast inference	EEG-only (requires hardware); no explainability; limited to depression, not suicide risk; no deployment/privacy	Our text-based approach + XAI (SHAP & Attention) + MLOps + on-device DPDP compliance is more accessible & practical
Hao Tang et al <i>arXiv preprint arXiv:2508.11265, 2023.</i>	XGBoost + SHAP for identifying key features in suicide risk prediction	Interpretable results using SHAP; clinician-friendly feature importance	Traditional ML (weaker on complex text); limited scalability & deployment	Advanced with fine-tuned BERT/MentalBERT + multiple XAI methods + complete MLOps + efficient on-device optimization under DPDP
Shaoxiong Ji et al.IEEE Access, vol. 10, pp. 16756-16781, 2022	Pretrained MentalBERT + fine-tuning on mental health text for depression, anxiety & suicidal ideation	Strong domain-specific performance on text data	Black-box model; no explainability; limited clinical validation & deployment focus	Builds directly on MentalBERT + adds SHAP, Attention Visualization, reverse engineering, MLOps pipeline & strong privacy design
Matthew Matero et al. (arXiv preprint arXiv:1611.09940, 2019)	Dual-context BERT framework using post content + user history for suicide risk	Captures longitudinal behavior for better context	Black-box; limited interpretability & deployment strategy	Enhanced with modern XAI techniques, full MLOps, and privacy-preserving on-device inference

Research Limitations:

- Limited Dataset Diversity – Most publicly available datasets are collected from social media platforms (e.g., Reddit), which may not represent all demographic and clinical populations.
- Black-Box Nature of Deep Learning Models – Although explainability techniques are used, they may not fully capture the complete reasoning behind model predictions.
- Limited Clinical Validation – The system requires extensive evaluation and feedback from mental health professionals before real-world clinical adoption.
- Language Dependency – The current model is primarily trained on English text and may not generalize well to other languages without retraining.
- Risk of Prediction Errors – False positives and false negatives may occur, making the model unsuitable as the sole basis for clinical decision-making.

References:

- [1] Muhammad Azhar et al. (2026)—Explainable Transformer-Based Framework for Suicide Risk Detection: Deep Learning with Interpretability for Mental Health Crisis Identification (2026)— MDPI.. Journals & Magazines, Information, Volume: 17
- [2]Pascal Penava et al.(2025)—Early-Stage Non-Severe Depression Detection Using a Novel CNNApproach on Resting-State EEG Data(2024)— IEEE. (<https://ieeexplore.ieee.org/document/10757418>). Journals & Magazines, IEEE Access, Volume: 12
- [3] A. Vázquez-Rodríguez, J. L. Hernández-Ramos, A. J. Jara, and A. F. Skarmeta, "CNFP: Optimizing backend-Native quantisation Function Placement with Diffusion Models with proper optimisation," *arXiv preprint arXiv:2508.11265*, 2024.
- [4] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive Recurrence Learning on Mental Health," *Advances in MLOps (Algorithms)*, Vol. 30, 2023.
- [5] I. Bello, H. Pham, Q. V. Le, M. Norouzi, and S. Bengio, "Neural Combinatorial Optimization with Reinforcement Learning," *arXiv preprint arXiv:1611.09940*, 2021.
- [6]P. Penava and R. Buettner, 'Early-Stage Non-Severe Depression Detection Using a Novel Convolutional Neural Network Approach Based on Resting-State EEG Data,' IEEE Access, vol. 12, pp. 173380-173389, 2024.
- [7] A. Dev et al., 'Exploration of EEG-based depression biomarkers identification techniques and their applications: A systematic review,' IEEE Access, vol. 10, pp. 16756-16781, 2022

Author & Year	Work Done	Advantages	Limitations	Comparison with Our Solution
Pascal Penava et al. (2024)	1D CNN on resting-state EEG for mild depression detection	Good accuracy (69.21%), low compute & fast inference	EEG-only (needs hardware); no explainability; not suicide-focused; no deployment/privacy	Our text-based + XAI + MLOps + on-device DPDP compliant approach is more accessible
Shaoxiong Ji et al. (2022)	Pretrained MentalBERT + fine-tuning on mental health text	Strong domain-specific performance	Black-box; no explainability or deployment focus	Builds on MentalBERT + adds SHAP, Attention, MLOps & privacy
Matthew Matero et al. (2019)	Dual-context BERT (post + history) for suicide risk	Better context via user history	Black-box; limited interpretability & deployment	Enhanced with modern XAI + production MLOps + on-device inference
Hao Tang et al. (2023)	XGBoost + SHAP for suicide risk features	Interpretable using SHAP	Traditional ML (weaker on text); poor deployment	Advanced with BERT + multiple XAI + full MLOps + on-device optimization

Conclusion:

This project presents an early-stage (approximately 30–40% complete) prototype for suicide risk prediction built on a BERT backbone with a basic attention-based explainability layer. The current build establishes the end-to-end pipeline — model architecture, inference flow, and a working FastAPI and Streamlit interface — but the classification head is untrained, so no clinical accuracy figure can be claimed yet. The reverse-engineering component completed so far is limited to simple attention-based token importance; deeper explainability methods such as SHAP, BertViz, and TCAV are planned but not yet implemented. Future work includes fine-tuning on labeled clinical datasets, completing the XAI and MLOps tooling (MLflow, Optuna), and optimizing the model for on-device deployment (ONNX/TFLite) with proper clinical validation before this can serve as a genuine decision-support tool.